Chapter 2

Chapter 2: Motion in One Dimension

- **As a first step in studying classical mechanics, we describe the motion of an object**

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Chapter 2: Part 1

Kinematics

- Describes motion while ignoring the agents that caused the motion
- For now, will consider motion in one dimension
  - Along a straight line (x-axis)
- Will use the particle model
  - A particle is a point-like object, has mass but infinitesimal size (i.e., the size is very small)

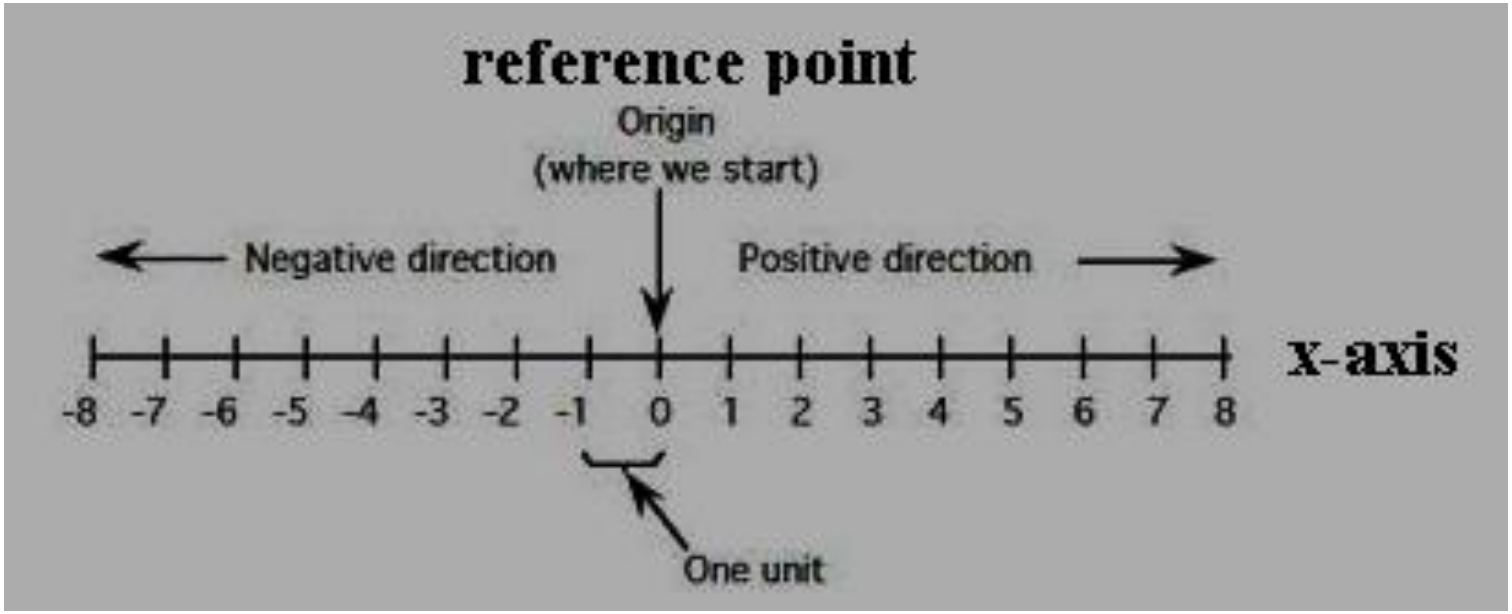
Vector and scalar quantities

- A **scalar quantity** **is completely specified by a single value with an appropriate unit and has no direction.**
  - examples of scalar quantities are volume, mass, speed, and time,….
  - Some scalars are always positive, such as mass and speed.
- A **vector quantity** **is completely specified by a number with an appropriate unit (the *magnitude of the vector) plus a direction.***
  - example of a vector quantity is displacement, velocity, and force,….

2.1 Position, Velocity, and Speed

Position (*x*):

- A particle's position x **is the location of the particle with respect to a chosen reference point** that we can consider to be the origin of a coordinate system.
- The motion of a particle is completely known if the particle's position in space is known at all times.



Displacement (Δx)

- The **displacement Δx *of a particle is defined as*** its change in position in some time interval.
- As the particle moves from an initial position x_i *to a final position* x_f *, its displacement is given by*

$$\Delta x = x_f - x_i \qquad (2.1)$$ ◀ Displacement

From this definition, we see that Δx *is positive if* x_f *is greater than xi and negative if* x_f *is* less than x_i.

Distance:

- It is very important to recognize the difference between displacement and distance traveled.
- **Distance is the length of a path followed by a particle.**
- Distance is always represented as a positive number, whereas displacement can be either positive or negative.
- Displacement is a vector quantity.
- Distance is a scalar quantity.

average velocity

- The **average velocity $v_{x,avg}$** ***of a particle is defined*** as the particle's displacement Δx *divided by the time interval* Δt *during which that* displacement occurs:

$$v_{x,avg} = \frac{\Delta x}{\Delta t} \qquad (2.2)$$ ◀ Average velocity

where the subscript *x indicates motion along the x axis. From this definition we see* that average velocity has dimensions of length divided by time (L/T), or meters per second in SI units.

Velocity is a vector quantity.

The average speed v_{avg}

- The **average speed** v_{avg} ***of a particle,*** ***a scalar*** **quantity**, is defined as the total distance ***d*** *traveled divided by the total time interval* required to travel that distance:

Average speed ▶ $$v_{avg} = \frac{d}{\Delta t} \qquad (2.3)$$

The SI unit of average speed is the same as the unit of average velocity: meters per second.

Example 2.1 Calculating the Average Velocity and Speed

- Find the displacement, average velocity, and average speed of the car in Figure 2.1a between positions A and F.

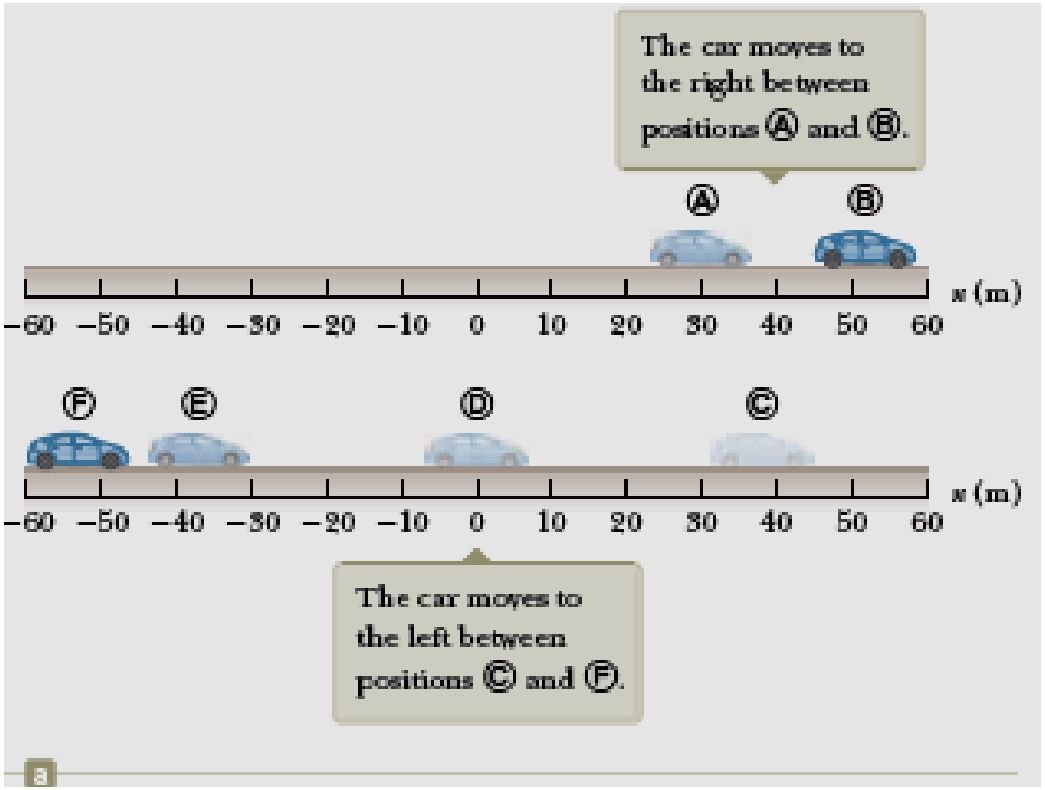


Table 2.1 Position of the Car at Various Times

Position	t (s)	x (m)
Ⓐ	0	30
Ⓑ	10	52
Ⓒ	20	38
Ⓓ	30	0
Ⓔ	40	−37
Ⓕ	50	−53

Solution:

Consult Figure 2.1 to form a mental image of the car and its motion. We model the car as a particle. From the position–time graph given in Figure 2.1b, notice that $x_{Ⓐ} = 30$ m at $t_{Ⓐ} = 0$ s and that $x_{Ⓑ} = -53$ m at $t_{Ⓑ} = 50$ s.

Use Equation 2.1 to find the displacement of the car:

$$\Delta x = x_{Ⓑ} - x_{Ⓐ} = -53\ \text{m} - 30\ \text{m} = -83\ \text{m}$$

This result means that the car ends up 83 m in the negative direction (to the left, in this case) from where it started. This number has the correct units and is of the same order of magnitude as the supplied data. A quick look at Figure 2.1a indicates that it is the correct answer.

Use Equation 2.2 to find the car's average velocity:

$$v_{x,\text{avg}} = \frac{x_{Ⓑ} - x_{Ⓐ}}{t_{Ⓑ} - t_{Ⓐ}}$$

$$= \frac{-53\ \text{m} - 30\ \text{m}}{50\ \text{s} - 0\ \text{s}} = \frac{-83\ \text{m}}{50\ \text{s}} = -1.7\ \text{m/s}$$

We cannot unambiguously find the average speed of the car from the data in Table 2.1 because we do not have information about the positions of the car between the data points. If we adopt the assumption that the details of the car's position are described by the curve in Figure 2.1b, the distance traveled is 22 m (from Ⓐ to Ⓑ) plus 105 m (from Ⓑ to Ⓕ), for a total of 127 m.

Use Equation 2.3 to find the car's average speed:

$$v_{\text{avg}} = \frac{127\ \text{m}}{50\ \text{s}} = 2.5\ \text{m/s}$$

Notice that the average speed is positive, as it must be. Suppose the red-brown curve in Figure 2.1b were different so that between 0 s and 10 s it went from Ⓐ up to 100 m and then came back down to Ⓑ. The average speed of the car would change because the distance is different, but the average velocity would not change.

2.2 Instantaneous Velocity and Speed

- describe an object's motion at any moment in time.
- the **instantaneous velocity v_x** equals the limiting value of the ratio ***$\Delta x/\Delta t$ as Δt approaches zero***

$$v_x = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} \qquad (2.4)$$

In calculus notation, this limit is called the *derivative* of x with respect to t, written dx/dt:

$$v_x = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt} \qquad (2.5)$$

The **instantaneous speed of a particle is defined as the magnitude of its instantaneous** velocity. As with average speed, instantaneous speed has no direction associated with it. For example, if one particle has an instantaneous velocity of +25 m/s along a given line and another particle has an instantaneous velocity of -25 m/s along the same line, both have a speed of 25 m/s.

Notice that:

- **Instantaneous Speed and Instantaneous Velocity**

- The magnitude of the average velocity **is not** the average speed.
- The magnitude of the instantaneous velocity, however, ***is*** *the instantaneous* speed.
- **In an infinitesimal time interval, the magnitude of the displacement is equal to the distance traveled by the particle.**

Conceptual Example 2.2 The Velocity of Different Objects

Consider the following one-dimensional motions: (**A**) a ball thrown directly upward rises to a highest point and falls back into the thrower's hand; (**B**) a race car starts from rest and speeds up to 100 m/s; and (**C**) a spacecraft drifts through space at constant velocity. Are there any points in the motion of these objects at which the instantaneous velocity has the same value as the average velocity over the entire motion? If so, identify the point(s).

SOLUTION

(A) The average velocity for the thrown ball is zero because the ball returns to the starting point; therefore, its displacement is zero. There is one point at which the instantaneous velocity is zero: at the top of the motion.

(B) The car's average velocity cannot be evaluated unambiguously with the information given, but it must have some value between 0 and 100 m/s. Because the car will have every instantaneous velocity between 0 and 100 m/s at some time during the interval, there must be some instant at which the instantaneous velocity is equal to the average velocity over the entire motion.

(C) Because the spacecraft's instantaneous velocity is constant, its instantaneous velocity at *any* time and its average velocity over *any* time interval are the same.

Example 2.3 Average and Instantaneous Velocity

A particle moves along the x axis. Its position varies with time according to the expression $x = -4t + 2t^2$, where x is in meters and t is in seconds.[3] The position–time graph for this motion is shown in Figure 2.4a. Because the position of the particle is given by a mathematical function, the motion of the particle is completely known, unlike that of the car in Figure 2.1. Notice that the particle moves in the negative x direction for the first second of motion, is momentarily at rest at the moment $t = 1$ s, and moves in the positive x direction at times $t > 1$ s.

(A) Determine the displacement of the particle in the time intervals $t = 0$ to $t = 1$ s and $t = 1$ s to $t = 3$ s.

In the first time interval, set $t_i = t_{@} = 0$ and $t_f = t_{@} = 1$ s and use Equation 2.1 to find the displacement:

$$\Delta x_{@\to@} = x_f - x_i = x_{@} - x_{@}$$
$$= [-4(1) + 2(1)^2] - [-4(0) + 2(0)^2] = -2 \text{ m}$$

For the second time interval ($t = 1$ s to $t = 3$ s), set $t_i = t_{@} = 1$ s and $t_f = t_{@} = 3$ s:

$$\Delta x_{@\to@} = x_f - x_i = x_{@} - x_{@}$$
$$= [-4(3) + 2(3)^2] - [-4(1) + 2(1)^2] = +8 \text{ m}$$

(B) Calculate the average velocity during these two time intervals.

In the first time interval, use Equation 2.2 with $\Delta t = t_f - t_i = t_{②} - t_{①} = 1$ s:

$$v_{x,avg(①\to②)} = \frac{\Delta x_{①\to②}}{\Delta t} = \frac{-2\text{ m}}{1\text{ s}} = -2\text{ m/s}$$

In the second time interval, $\Delta t = 2$ s:

$$v_{x,avg(②\to③)} = \frac{\Delta x_{②\to③}}{\Delta t} = \frac{8\text{ m}}{2\text{ s}} = +4\text{ m/s}$$

(C) Find the instantaneous velocity of the particle at $t = 2.5$ s.

$$x = -4t + 2t^2$$

$$\frac{dx}{dt} = -4 + 4t \text{ so at } t = 2.5\text{ s so } \frac{dx}{dt} = -4 + 4(2.5) = +6\frac{m}{s}$$

$$or\ v = \lim_{\Delta t=0}\frac{\Delta x}{\Delta t} = \lim_{\Delta t=0}\frac{x(t+\Delta t) - x(t)}{\Delta t} = \frac{-4(t+\Delta t) + 2(t+\Delta t)^2 - (-4t + 2t^2)}{\Delta t}$$

$$= \lim_{\Delta t=0}\frac{-4t - 4\Delta t + 2(t^2 + \Delta t^2 + 2t\Delta t) + 4t - 2t^2}{\Delta t} = \lim_{\Delta t=0} -4 + \Delta t + 2t$$

$$= -4 + 2t \text{ so at } t = 2.5\text{ s so } v = -4 + 10 = +6\frac{m}{s}$$

Chapter 2: Part 2

2.3 Analysis Model: Particle Under Constant Velocity

- **analysis model is a common situation that occurs time and again when solving** physics problems.
- **Position as a function of time for the particle under constant velocity model**

$$x_f = x_i + v\Delta t \; as \; t_i = 0 \; and \; t_f = t \; \rightarrow$$

$$\boldsymbol{x_f = x_i + vt}$$

Example 2.4 Modeling a Runner as a Particle

A kinesiologist is studying the biomechanics of the human body. (*Kinesiology* is the study of the movement of the human body. Notice the connection to the word *kinematics*.) She determines the velocity of an experimental subject while he runs along a straight line at a constant rate. The kinesiologist starts the stopwatch at the moment the runner passes a given point and stops it after the runner has passed another point 20 m away. The time interval indicated on the stopwatch is 4.0 s.

(A) What is the runner's velocity?

SOLUTION

We model the moving runner as a particle because the size of the runner and the movement of arms and legs are unnecessary details. Because the problem states that the subject runs at a constant rate, we can model him as a *particle under constant velocity*.

Having identified the model, we can use Equation 2.6 to find the constant velocity of the runner:

$$v_x = \frac{\Delta x}{\Delta t} = \frac{x_f - x_i}{\Delta t} = \frac{20\text{ m} - 0}{4.0\text{ s}} = 5.0\text{ m/s}$$

(B) If the runner continues his motion after the stopwatch is stopped, what is his position after 10 s have passed?

SOLUTION

Use Equation 2.7 and the velocity found in part (A) to find the position of the particle at time $t = 10$ s:

$$x_f = x_i + v_x t = 0 + (5.0\text{ m/s})(10\text{ s}) = 50\text{ m}$$

Is the result for part (A) a reasonable speed for a human? How does it compare to world-record speeds in 100-m and 200-m sprints? Notice the value in part (B) is more than twice that of the 20-m position at which the stopwatch was stopped. Is this value consistent with the time of 10 s being more than twice the time of 4.0 s?

A particle moving with a constant speed through a distance *d* along a curved path.

- As an example, imagine a particle moving at a constant speed in a circular path. If the speed is 5.00 m/s and the radius of the path is 10.0 m, we can calculate the time interval required to complete one trip around the circle:

$$v = \frac{d}{\Delta t} \quad \rightarrow \quad \Delta t = \frac{d}{v} = \frac{2\pi r}{v} = \frac{2\pi(10.0\ \mathrm{m})}{5.00\ \mathrm{m/s}} = 12.6\ \mathrm{s}$$

2.4 Acceleration

- The **average acceleration $a_{x,avg}$ *of the particle is defined as the change*** in velocity Δv_x *divided by the time interval* Δt *during which that change occurs:*

$$a_{x,avg} = \frac{\Delta v_x}{\Delta t} = \frac{v_{xf} - v_{xi}}{t_f - t_i} \qquad (2.9)$$

Average acceleration

The SI unit of acceleration is meters per second squared (m/s^2).

The instantaneous acceleration:

$$a_x = \lim_{\Delta t \to 0} \frac{\Delta v_x}{\Delta t} = \frac{dv_x}{dt} \qquad (2.10)$$

in one-dimensional motion, the acceleration equals the *second derivative of x with respect to time.*

Because $v_x = dx/dt$, the acceleration can also be written as

$$a_x = \frac{dv_x}{dt} = \frac{d}{dt}\left(\frac{dx}{dt}\right) = \frac{d^2x}{dt^2} \qquad (2.12)$$

Graphical Relationships Between x, v_x, and a_x

- The average velocity is $v\ (average) = \frac{\Delta x}{\Delta t} = \frac{x(t_f) - x(t_i)}{t_f - t_i}$
- The velocity at any instant is the slope of the tangent to the ***x–t graph*** *at that instant.*
- The average acceleration is $a\ (average) = \frac{\Delta v}{\Delta t} = \frac{v(t_f) - v(t_i)}{t_f - t_i}$
- The acceleration at any instant is the slope of the tangent to the ***v_x–t graph*** *at that instant.*

See Conceptual Example 2.5 page 33

Example 2.6 Average and Instantaneous Acceleration

The velocity of a particle moving along the x axis varies according to the expression $v_x = 40 - 5t^2$, where v_x is in meters per second and t is in seconds.

(A) Find the average acceleration in the time interval $t = 0$ to $t = 2.0$ s.

Find the velocities at $t_i = t_{@} = 0$ and $t_f = t_{@} = 2.0$ s by substituting these values of t into the expression for the velocity:

$$v_{x@} = 40 - 5t_{@}^2 = 40 - 5(0)^2 = +40 \text{ m/s}$$

$$v_{x@} = 40 - 5t_{@}^2 = 40 - 5(2.0)^2 = +20 \text{ m/s}$$

Find the average acceleration in the specified time interval $\Delta t = t_{@} - t_{@} = 2.0$ s:

$$a_{x\text{avg}} = \frac{v_{xf} - v_{xi}}{t_f - t_i} = \frac{v_{x@} - v_{x@}}{t_{@} - t_{@}} = \frac{20 \text{ m/s} - 40 \text{ m/s}}{2.0 \text{ s} - 0 \text{ s}}$$

$$= -10 \text{ m/s}^2$$

(B) Determine the acceleration at $t = 2.0$ s.

$$a = \frac{dv}{dt} = -10t = -10 * 2 = -20 \frac{m}{s^2}$$

$$v_{xf} = 40 - 5(t + \Delta t)^2 = 40 - 5t^2 - 10t\,\Delta t - 5(\Delta t)^2$$

$$\Delta v_x = v_{xf} - v_{xi} = -10t\,\Delta t - 5(\Delta t)^2$$

$$a_x = \lim_{\Delta t \to 0} \frac{\Delta v_x}{\Delta t} = \lim_{\Delta t \to 0} (-10t - 5\,\Delta t) = -10t$$

$$a_x = (-10)(2.0) \text{ m/s}^2 = -20 \text{ m/s}^2$$

2.6 Analysis Model: Particle Under Constant Acceleration

- **kinematic equations** that may be used to solve any problem involving a particle **under constant acceleration** in one dimension.
- In such a case, the average acceleration $a_{x,avg}$ *over any time interval is numerically* ***equal to*** *the instantaneous acceleration* a_x at any instant within the interval, and the velocity changes at the same rate throughout the motion.
- We generate several equations that describe the motion of a particle for this model.

$$v_{xf} = v_{xi} + a_x t \quad \text{(for constant } a_x\text{)} \qquad (2.13)$$

Because velocity at constant acceleration varies linearly in time according to Equation 2.13, we can express the average velocity in any time interval as the arithmetic mean of the initial velocity v_{xi} *and the final velocity* v_{xf} *:*

$$v_{x,avg} = \frac{v_{xi} + v_{xf}}{2} \quad \text{(for constant } a_x\text{)} \qquad (2.14)$$

Chapter 2: Part 3

- **Position as a function of velocity and time for the particle under constant acceleration model**

$$x_f = x_i + \tfrac{1}{2}(v_{xi} + v_{xf})t \quad \text{(for constant } a_x\text{)} \qquad (2.15)$$

$$x_f = x_i + v_{xi}t + \tfrac{1}{2}a_x t^2 \quad \text{(for constant } a_x\text{)} \qquad (2.16)$$

- **Velocity as a function of position for the particle under constant acceleration model**

$$v_{xf}^2 = v_{xi}^2 + 2a_x(x_f - x_i) \quad \text{(for constant } a_x\text{)} \qquad (2.17)$$

For motion at *zero* acceleration, we see from Equations 2.13 and 2.16 that

$$\left.\begin{aligned} v_{xf} &= v_{xi} = v_x \\ x_f &= x_i + v_x t \end{aligned}\right\} \quad \text{when } a_x = 0$$

That is, when the acceleration of a particle is zero, its velocity is constant and its position changes linearly with time. In terms of models, when the acceleration of a particle is zero, the particle under constant acceleration model reduces to the particle under constant velocity model (Section 2.3).

kinematic equations: Particle Under Constant Acceleration

$$v_{xf} = v_{xi} + a_x t \tag{2.13}$$

$$v_{x,\text{avg}} = \frac{v_{xi} + v_{xf}}{2} \tag{2.14}$$

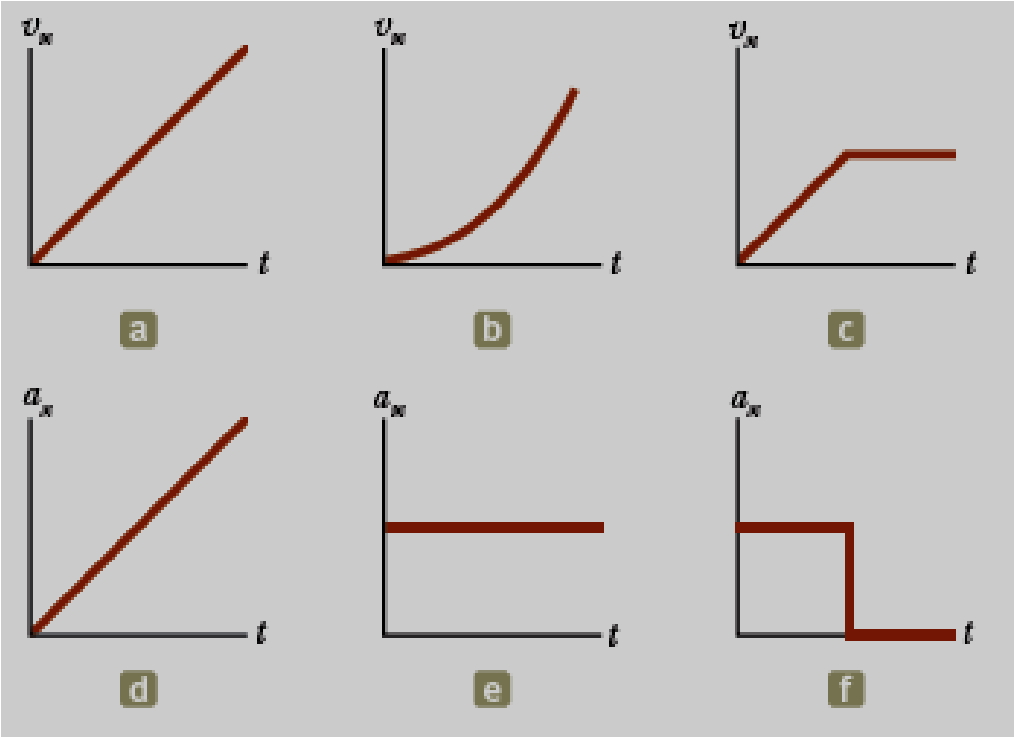
$$x_f = x_i + \tfrac{1}{2}(v_{xi} + v_{xf})t \tag{2.15}$$

$$x_f = x_i + v_{xi}t + \tfrac{1}{2}a_x t^2 \tag{2.16}$$

$$v_{xf}^2 = v_{xi}^2 + 2a_x(x_f - x_i) \tag{2.17}$$

Quick Quiz 2.6 In Figure 2.12, match each *v_x–t graph on the top with the a_x–t* graph on the bottom that best describes the motion. (page 38)

- **Figure 2.12 (Quick Quiz 2.6):** Parts (a), (b), and (c) are *v_x–t graphs* of objects in one-dimensional motion. The possible accelerations of each object as a function of time are shown in scrambled order in (d), (e), and (f).



Example 2.7 Carrier Landing

A jet lands on an aircraft carrier at a speed of 140 mi/h ($\approx$ 63 m/s).

(A) What is its acceleration (assumed constant) if it stops in 2.0 s due to an arresting cable that snags the jet and brings it to a stop?

Because the acceleration of the jet is assumed constant, we model it as ***a particle under constant acceleration***.

Equation 2.13 is the only equation in the particle under constant acceleration model that does not involve position, so we use it to find the acceleration of the jet, modeled as a particle:

$$a_x = \frac{v_{xf} - v_{xi}}{t} \approx \frac{0 - 63\ \text{m/s}}{2.0\ \text{s}}$$

$$= -32\ \text{m/s}^2$$

(B) If the jet touches down at position $x_i = 0$, what is its final position?

Use Equation 2.15 to solve for the final position:

$$x_f = x_i + \tfrac{1}{2}(v_{xi} + v_{xf})t = 0 + \tfrac{1}{2}(63\ \text{m/s} + 0)(2.0\ \text{s}) = 63\ \text{m}$$

The idea of using arresting cables to slow down landing aircraft and enable them to land safely on ships

Example 2.8 Watch Out for the Speed Limit!

A car traveling at a constant speed of 45.0 m/s passes a trooper on a motorcycle hidden behind a billboard. One second after the speeding car passes the billboard, the trooper sets out from the billboard to catch the car, accelerating at a constant rate of 3.00 m/s^2. How long does it take the trooper to overtake the car?

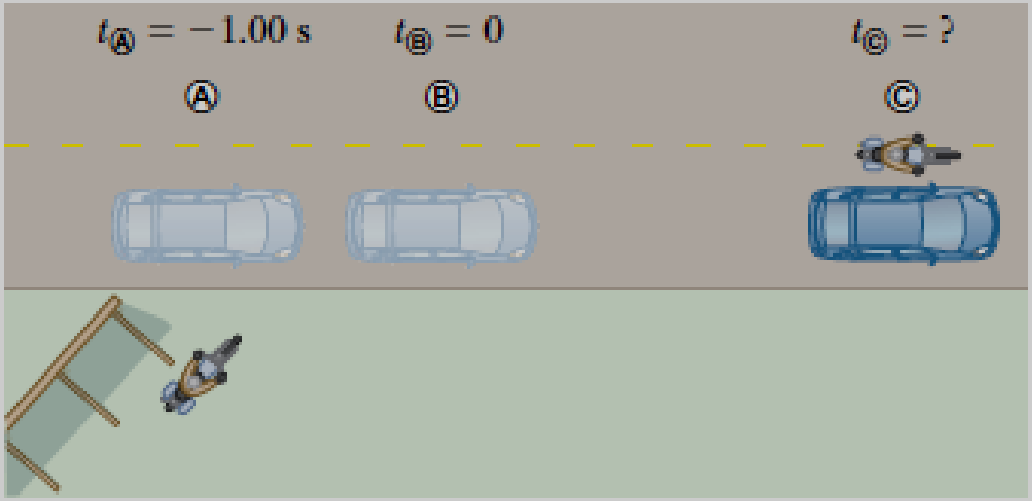


Figure 2.13 (Example 2.8) A speeding car passes a hidden trooper.

The car is modeled as a *particle under* ***constant velocity***, *and* ***the trooper*** *is modeled as a particle under* ***constant acceleration.***

Key solution:

$x_{trooper} = x_{car}$

At that instant, the car has already traveled a distance of 45.0 m from the billboard because it has traveled at a constant speed of $v_x = 45.0$ m/s for 1 s. Therefore, the initial position of the speeding car is $x_B = 45.0$ m.

Using the particle under constant velocity model, apply Equation 2.7 to give the car's position at any time t:

$$x_{car} = x_B + v_{x\,car}t$$

A quick check shows that at $t = 0$, this expression gives the car's correct initial position when the trooper begins to move: $x_{car} = x_B = 45.0$ m.

The trooper starts from rest at $t_B = 0$ and accelerates at $a_x = 3.00\ \text{m/s}^2$ away from the origin. Use Equation 2.16 to give her position at any time t:

$$x_f = x_i + v_{xi}t + \tfrac{1}{2}a_x t^2$$
$$x_{trooper} = 0 + (0)t + \tfrac{1}{2}a_x t^2 = \tfrac{1}{2}a_x t^2$$

Set the positions of the car and trooper equal to represent the trooper overtaking the car at position C:

$$x_{trooper} = x_{car}$$
$$\tfrac{1}{2}a_x t^2 = x_B + v_{x\,car}t$$

Rearrange to give a quadratic equation:

$$\tfrac{1}{2}a_x t^2 - v_{x\,car}t - x_B = 0$$

Solve the quadratic equation for the time at which the trooper catches the car (for help in solving quadratic equations, see Appendix B.2.):

$$t = \frac{v_{x\,car} \pm \sqrt{v_{x\,car}^2 + 2a_x x_B}}{a_x}$$

$$(1)\quad t = \frac{v_{x\,car}}{a_x} \pm \sqrt{\frac{v_{x\,car}^2}{a_x^2} + \frac{2x_B}{a_x}}$$

Evaluate the solution, choosing the positive root because that is the only choice consistent with a time $t > 0$:

$$t = \frac{45.0\ \text{m/s}}{3.00\ \text{m/s}^2} + \sqrt{\frac{(45.0\ \text{m/s})^2}{(3.00\ \text{m/s}^2)^2} + \frac{2(45.0\ \text{m})}{3.00\ \text{m/s}^2}} = 31.0\ \text{s}$$

2.7 Freely Falling Objects

- It is well known that, in the absence of air resistance, all objects dropped near the Earth's surface fall toward the Earth with the same constant acceleration under the influence of the Earth's gravity.
- A freely falling object is any object moving freely under the influence of gravity alone, regardless of its initial motion.
- Objects thrown upward or downward and those released from rest are all falling freely once they are released. Any freely falling object experiences an acceleration directed *downward,* regardless of its initial motion.
- We shall denote the magnitude of the *free-fall acceleration, also called the acceleration due to gravity,* ***by the symbol g****.*
- At the Earth's surface, the value of *g is approximately 9.80 m/s^2.*
- **the particle under constant acceleration model can be applied**. The only modification for freely falling objects that we need to make in these equations is to note that the motion is in the vertical direction (the *y direction) rather than in the horizontal direction* (*x) and that the acceleration is downward and has a magnitude of 9.80 m/s^2.* Therefore, **we choose $a_y = -g = -9.80$ m/s^2**, *where the negative sign means that* the acceleration of a freely falling object is downward.

Example 2.10 Not a Bad Throw for a Rookie! AM

A stone thrown from the top of a building is given an initial velocity of 20.0 m/s straight upward. The stone is launched 50.0 m above the ground, and the stone just misses the edge of the roof on its way down as shown in Figure 2.14.

(A) Using $t_{Ⓐ} = 0$ as the time the stone leaves the thrower's hand at position Ⓐ, determine the time at which the stone reaches its maximum height.

$$v_f = v_i - gt$$

$$v_{yf} = v_{yi} + a_y t \quad \rightarrow \quad t = \frac{v_{yf} - v_{yi}}{a_y}$$

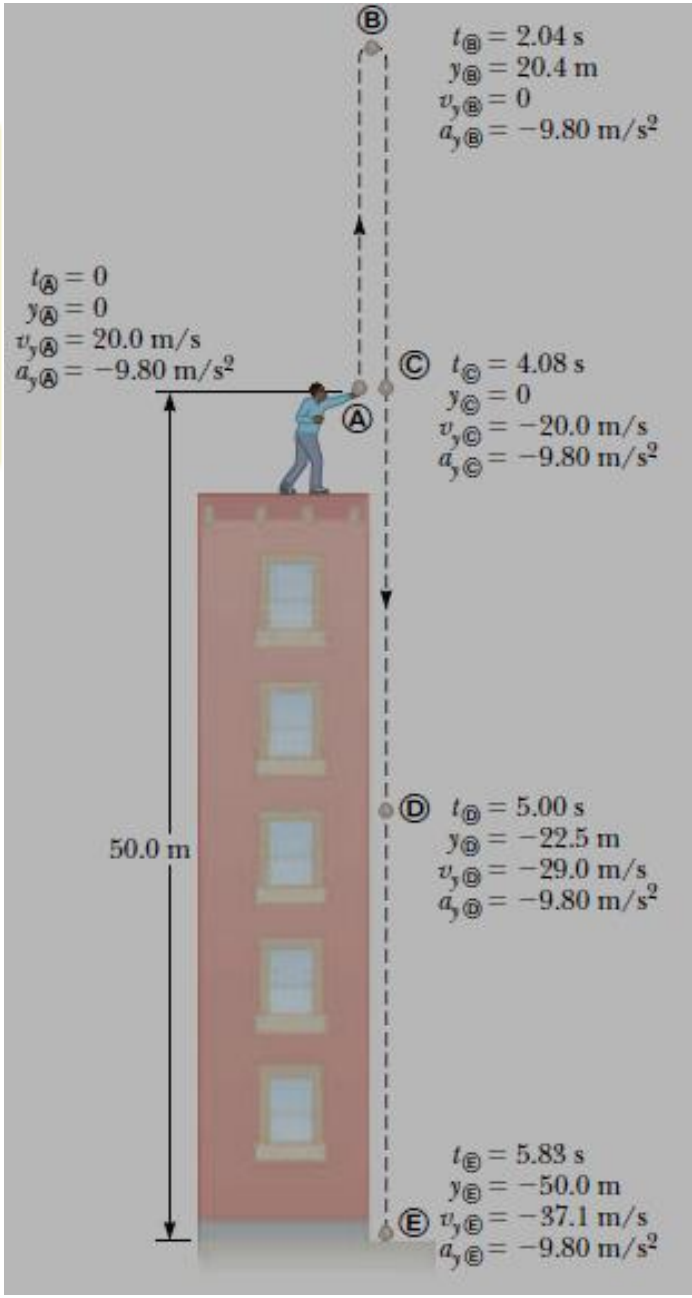
$$t = t_{Ⓑ} = \frac{0 - 20.0\ \text{m/s}}{-9.80\ \text{m/s}^2} = 2.04\ \text{s}$$

(B) Find the maximum height of the stone.

$$y_f = y_i + v_i t - \frac{1}{2} g t^2$$

$$y_{max} = y_{Ⓑ} = y_{Ⓐ} + v_{x Ⓐ} t + \tfrac{1}{2} a_y t^2$$

$$y_{Ⓑ} = 0 + (20.0\ \text{m/s})(2.04\ \text{s}) + \tfrac{1}{2}(-9.80\ \text{m/s}^2)(2.04\ \text{s})^2 = 20.4\ \text{m}$$



(C) Determine the velocity of the stone when it returns to the height from which it was thrown.

$$v_f^2 = v_i^2 - 2g\Delta y$$

$$v_{y©}^2 = v_{y@}^2 + 2a_y(y_© - y_@)$$

$$v_{y©}^2 = (20.0\ \text{m/s})^2 + 2(-9.80\ \text{m/s}^2)(0 - 0) = 400\ \text{m}^2/\text{s}^2$$

$$v_{y©} = -20.0\ \text{m/s}$$

(D) Find the velocity and position of the stone at $t = 5.00$ s.

$$v_{y®} = v_{y@} + a_y t = 20.0\ \text{m/s} + (-9.80\ \text{m/s}^2)(5.00\ \text{s}) = -29.0\ \text{m/s}$$

$$y_® = y_@ + v_{y@}\,t + \tfrac{1}{2}a_y t^2$$

$$= 0 + (20.0\ \text{m/s})(5.00\ \text{s}) + \tfrac{1}{2}(-9.80\ \text{m/s}^2)(5.00\ \text{s})^2$$

$$= -22.5\ \text{m}$$

2.8 Kinematic Equations Derived from Calculus

Velocity versus time graph for a particle moving along the *x axis.* ***The total area under the*** **curve is the total displacement of the particle.**

$$v_{avg} = \frac{\Delta x}{\Delta t} \rightarrow \Delta x = v_{avg}\Delta t$$

$$v_{avg} = \frac{dx}{dt} \rightarrow dx = v_{avg}dt \rightarrow \Delta x = \int_{t_i}^{t_f} v(t)\,dt$$

acceleration versus time graph for a particle moving along the *x axis.* ***The total area under the*** **curve is the delta velocity of the particle.**

$$\boldsymbol{a_{avg} = \frac{\Delta v}{\Delta t} \rightarrow \Delta v = a_{avg}\Delta t = v_f - v_i}$$

Chapter 2: Part 4
Problems with solutions